Introduction

This report presents an evaluation of a subset of the Open ASR Leaderboard. It contains analyses of reproducibility, normalization, and a proposed metric adapted toward critical words.

Resource selection

Leaderboard selection

I chose to work with the Open ASR Leaderboard[1] available on Hugging Face. This leaderboard is entirely open source and aims to be reproducible, with its datasets, models, and scoring pipelines being publicly available. This transparency should help scientists understand exactly how the leaderboard's published numbers were obtained.

This leaderboard also evaluates performance on a large variety of datasets and is a regularly updating collaboration between industry, academia, and the open-source community (Hugging Face, NVIDIA, Cambridge, Mistral, OpenAI), so it represents a range of scientifically and practically interesting use cases.

Model selection

From this leaderboard I selected models to work with from three families: Whisper, Qwen, and Granite. I chose these families and specific models because each represents a different interesting part of the ASR landscape. Whisper Large V3 Turbo is a widely recognized baseline in the community, Qwen3-ASR-1.7B has known high performance on Word Error Rate (WER, highest on the leaderboard), and Granite TurboCTC-nc is known for being fast in terms of RTF (Real Time Factor, highest on the leaderboard in RTFx, the reciprocal of RTF). Table 1 shows the position of these models on the leaderboard across all datasets.

Model	Architecture	Rank	Avg WER	Avg RTFx
Qwen3-ASR-1.7B-hf	LLM autoregressive	9	5.40	819.30
Granite TurboCTC-nc (470M)	Pure CTC	24	6.02	20831.46
Whisper Large V3 Turbo (809M)	Encoder-decoder	44	7.64	794.31

Table 1. Selected models from the Open ASR Leaderboard (57 models, updated October 22, 2026; averages conducted by leaderboard).

[1] Leaderboard: https://huggingface.co/spaces/hf-audio/open_asr_leaderboard
Published paper on the leaderboard: https://arxiv.org/html/2510.06961v4

Dataset selection

The Open ASR leaderboard evaluates 12 datasets comprising about 297 hours of audio in total. This size is prohibitive to run in a timely and cost-effective way on my available personal hardware (Mac Mini, Apple M4, 16GB RAM), so I aimed to choose a subset of this total that is representative of the real-life conditions that an ASR system might encounter. Ensuring sufficient representation of those conditions is more important for evaluating ASR robustness than ensuring proportional representation of the leaderboard's distribution of data types.

For this reason, I chose URGENT24 as the principal dataset for this report's experiments. This dataset of 1000 samples is the most representative of the variety of acoustic conditions found in real life, like background noise and echo, in addition to other conditions I characterized (below).

URGENT24 acoustic characterization

F0 analysis

Good ASR should perform well across a variety of speaker voices, so a good dataset should represent that variety. To characterize URGENT24 for this, I developed an F0 (fundamental frequency) estimation method and ran it on each audio file. F0 analysis allows speakers' vocal registers to be distinguished, with male voices typically being situated between 85-180 Hz and female voices between 165-255 Hz.

The method proceeds this way:

1. Divide the audio into frames of 40 ms with a stride of 10 ms.
2. For each frame, take out the average, apply a Hanning window, and calculate the autocorrelation.
3. Use parabolic interpolation to identify and refine the peak of the autocorrelation corresponding to a frequency between 60-400 Hz.
4. Retain as voiced only the frames with a sufficient threshold (peak confidence above 0.30).
5. Use the median F0 from the set of voiced frames to classify the speaker's vocal register.

Figure 1 shows that URGENT24 contains a higher proportion of male-range voices than female-range voices (57.6% vs. 35.7%). An intermediate zone (5.8%) falls between the two pitch regimes (F0), where the audio clips could correspond to one register or the other. While the distribution is not equal, the proportion of female-range voices is sufficiently large to validate ASR performance on both registers.

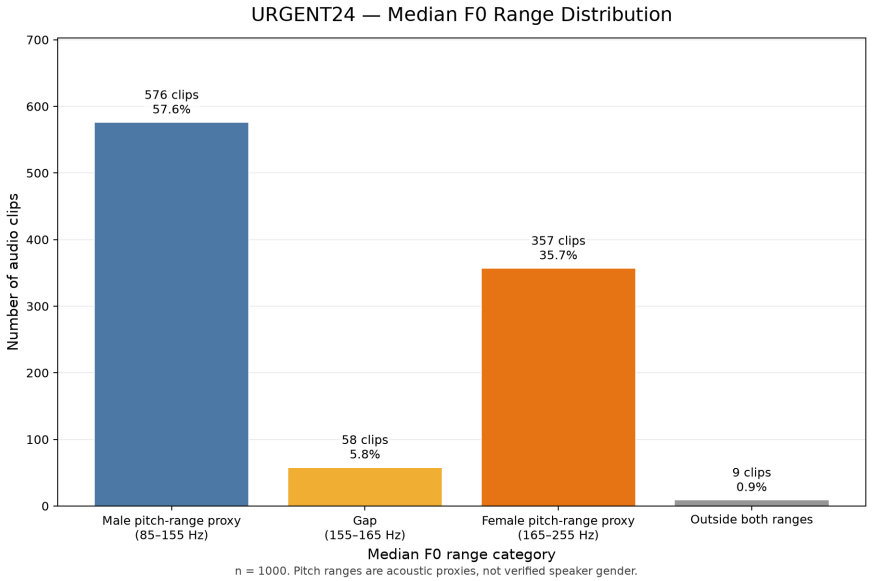


Figure 1. Distribution of median F0 ranges across 1,000 URGENT24 audio clips.

SNR analysis

Good ASR should also perform well in a variety of noisy conditions. URGENT24 contains in its metadata a measurement of the SNR (signal-to-noise ratio) for each sample, covering a generally uniform distribution ranging from -5 to 20 dB (Figure 2, left). This diversity of noise conditions is representative of a variety of real situations, from noisy ones (with negative SNR) to relatively clean ones (with high SNR).

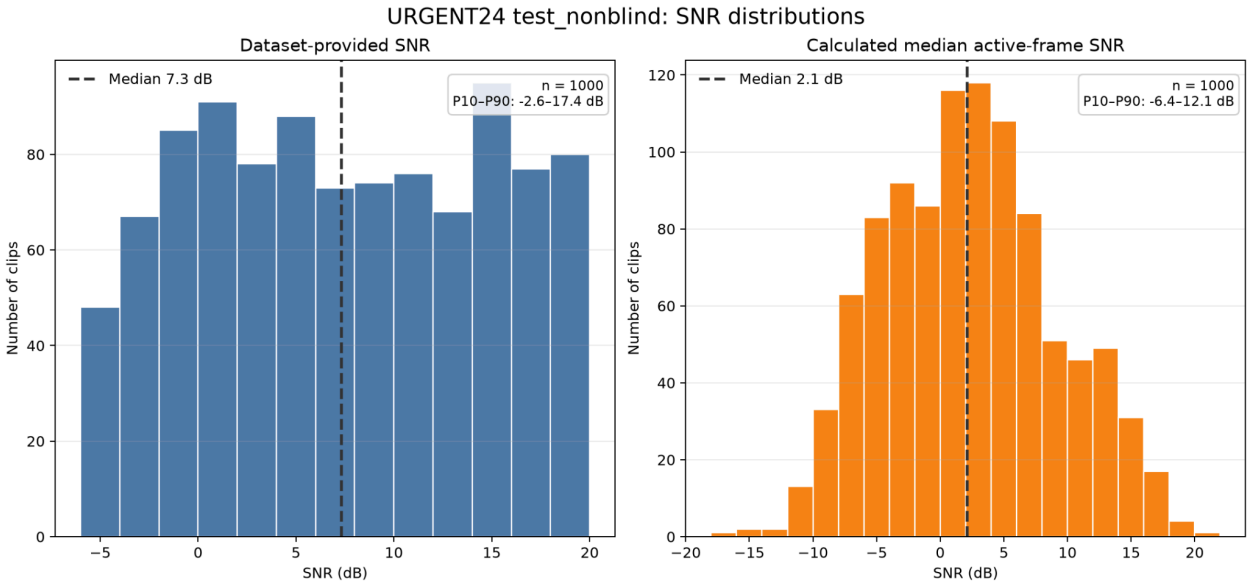


Figure 2. Distribution of dataset-provided vs. calculated SNR values across 1,000 URGENT24 audio clips.

However, the dataset's documentation did not detail how their SNRs were calculated. To get a clearer view of this, I recalculated the SNR directly using a method I know to be precise. The method estimates the noise based on the difference between the noisy audio and the clean audio (both provided by the dataset), then calculates the SNR frame by frame (20-ms frames, 10-ms intervals) on active frames only, taking the median per clip. The resulting SNR distribution (Figure 2, right) is different, with a more concentrated shape and a median of 2.1 dB, significantly lower than the 7.3 dB reported in the metadata. This difference, measured by a frame-level, high-resolution method, indicates that the dataset is noisier than could be measured with a more general or low-resolution method.

Reproduction of WER results and normalization analysis

Choosing URGENT24 as a subset (at 1.92 hours) also allowed timely and iterative experimentation on my available hardware, and confident validation of an evaluation pipeline aiming to reproduce the leaderboard's WER results.

First, I built a pipeline that simply calls Hugging Face inference APIs on URGENT24 using the three models, then scores using a standard implementation of WER. Performance was poor because the text was un-normalized before WER calculation (e.g., references and hypotheses with different casing, separated punctuation, etc.).

Then, I aimed to reuse the text normalization used across the leaderboard, which is the Whisper English normalizer, to reduce the gap. This normalizer is very mature, containing routines for case conversion, punctuation removal, filler word removal, contraction management, number conversion, and spelling standardization implemented by regex, manual rewrites, and other methods particular to English edge cases. It is recent and actively maintained and improved by the community. It is very difficult to improve on such a mature normalizer, so I did not reimplement these rules from scratch. Using this normalizer closed much of the WER gap.

Finally, while examining model outputs, I noticed that some hypotheses degenerated into repetition, exploding insertion errors. While the leaderboard documentation mentioned repetition suppression as part of their normalizer, I did not find details on implementation. I implemented repetition suppression myself, which closed even more of the WER gap.

Table 2 compares the leaderboard's WER with that of my local evaluation pipeline on URGENT24, over a variety of normalization conditions. With full normalization, results are similar enough to validate the pipeline. Remaining small discrepancies are likely due to numeric precision differences (related to local hardware vs. upstream GPU calculations), potentially different random seed effects, and differences in edge case normalization methods.

Model	Leaderboard WER	Local WER	Local WER + normalization	Local WER + normalization - repetition
Whisper Large V3 Turbo	13.69%	33.65%	22.55%	13.03%
Qwen3-ASR 1.7B	9.93%	24.72%	9.94%	9.94%
Granite Speech 5.0 TurboCTC	10.22%	29.46%	10.31%	10.31%

Table 2. Comparison of WER on URGENT24 between leaderboard values and local evaluation.

Audio normalization

Beyond text normalization, I also experimented with a form of normalization on the audio signal before input to the model for inference. I applied a band pass filter (70-7000 Hz) to see if attenuating low-frequency noise (due to e.g. vibrations or wind) and high-frequency noise (due to e.g. breathing) could improve WER.

Table 3 shows the results on URGENT24, which are mixed. The filter slightly improved Qwen, but degraded Whisper and Granite. This indicates that the filter also attenuated some useful components of the speech signal, and each model reacts differently depending on its training and architecture. This quick experiment shows that the approach has potential, but would need more fine adjustment of which frequencies to cut according to the type of targeted noise and model.

Model	WER (no filter)	WER (band pass 70–7000 Hz)	Δ
Qwen3-ASR 1.7B	9.94%	9.64%	−0.30 pp
Whisper Large V3 Turbo	13.03%	13.46%	+0.43 pp
Granite Speech 5.0 TurboCTC	10.31%	10.62%	+0.31 pp

Table 3. Effect of band pass filtering (70-7000 Hz) on WER for URGENT24.

WER by acoustic category

Scoring with full text normalization and without audio normalization, I also broke down WER performance using Whisper Large V3 Turbo by reporting into the acoustic categories previously discussed (Figures 3, 4).

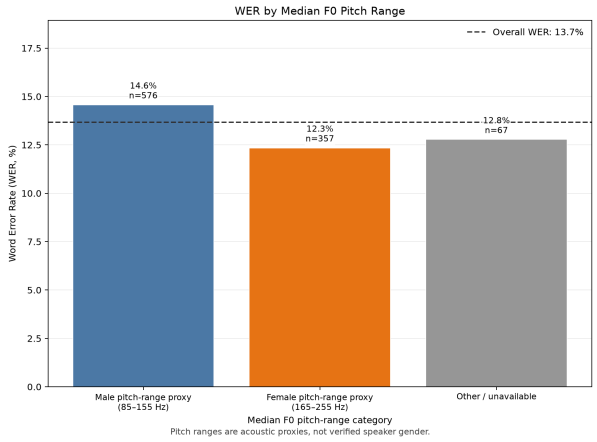


Figure 3. WER by median F0 pitch range, grouped into male-, female-, and other-range registers.

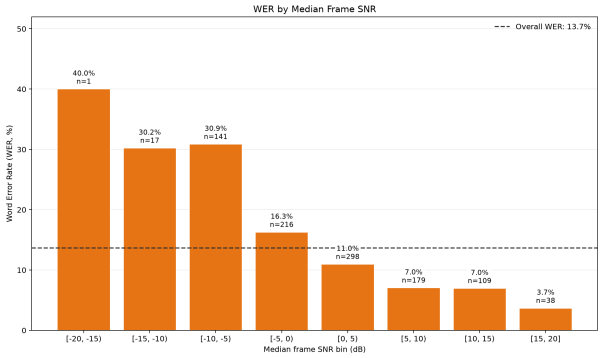


Figure 4. WER by median frame signal-to-noise ratio (SNR).

WER across speaker registers is fairly stable, with male-range voices performing slightly better. WER also increases with decreasing SNR, which is expected, given that noisier conditions are more difficult for the model. For negative SNRs (more noise), WER nearly doubles, suggesting the Whisper model is not sufficiently trained to be robust to noise.

New metric: Proper-name-weighted WER

Standard WER, as a metric, treats all errors as equal, whether on unremarkable words or proper names. In certain critical contexts, an error on certain words might have more significant consequences. For example, in the banking domain, transcribing the exact name of a customer might be essential, and in the medical domain, the written name of a medication might need to be exactly correct to avoid all confusion that could affect patient safety.

To address this kind of problem, I tested a variant of standard WER that I call *proper-name-weighted WER*. This metric assigns a higher weight (by default a factor of 3) to words identified as proper names (provideable by a list) while calculating WER. This will cause an error on a proper name to cost three times more than an error on an ordinary word, both in the numerator (cost of the error) and the denominator (the weighted number of reference words).

For example, given a reference sentence “Alice met Bob today” and a hypothesis sentence “Alicia met Bob today,” and assuming “Alice” is in the proper names list (weight = 3), the error on “Alice”→ “Alicia” raises the WER from 25% to 50%:

Metric	Calculation	Result
Standard WER	1 error / 4 words	25.0%
Proper-name-weighted WER	3 (weighted cost) / 6 (3+1+1+1)	50.0%

This metric informs the choice of ASR model for applications where certain terms absolutely must not be wrongly transcribed. For this experiment, I directed GitHub Copilot to read the URGENT24 references and construct a list of proper names they contained, though future work could do this in a more principled way.

Table 4 shows that the proper-name-weighted WER is consistently higher than the standard WER for all three models, with an increase ranging from 0.60 to 0.68 percentage points. This confirms that errors involving proper nouns contribute proportionally more to the overall error rate. The difference is relatively consistent across the three models, suggesting that none of them handles proper nouns significantly better than the others.

Model	Standard WER	Proper-name-weighted WER	Δ
Whisper large-v3-turbo	13.03%	13.71%	+0.68pp
Qwen3-ASR 1.7B	9.94%	10.54%	+0.60pp
Granite Speech 5.0 TurboCTC	10.62%	11.23%	+0.61pp

Table 4. Standard WER vs proper-name-weighted WER on URGENT24.

Further Comments

Limitations

While URGENT24 is large enough to be representative of a variety of conditions, this evaluation still is limited to its 1000 samples. These results may not necessarily be generalizable to very different corpora, languages, or audio conditions.

Even on URGENT24, local vs. leaderboard WER are not perfectly equivalent, since the hardware and resources used locally are different from those used to generate the leaderboard.

Proper-name-weighted WER depends entirely on the correctness of the provided list of proper names and the calibration of the chosen weight. This metric can also add some latency to the evaluation process, especially if the process for producing the list is costly.

Though F0 analysis is a well-established domain and can give a good indicator, pitch regimes related to speaker register are only acoustic estimations and do not allow us to confirm the real gender of a speaker.

Evaluation automation

The local evaluation pipeline is already largely automated, containing one routine for transcribing dataset audio (by model inference) and another for calculating WER scoring. To automate even more, these routines could be chained together so that scoring launches automatically at the end of transcription. These scripts include flags for easy calculation and comparison of different conditions (proper-name-weighted WER, text normalization, audio normalization, etc.) without modifying any code.

Use of AI tools

I used VSCode with GitHub Copilot to accelerate my implementation and to iterate on experiments faster. While in this loop, Copilot was helpful in enabling me to refine my ideas and quickly integrate new functionality to try. However, the thinking is all mine, including pipeline design, scoring ideas, normalization methods, etc. I also used Claude AI to help me formulate a presentation structure and some coherent phrasing for this report.